An absolute-norming lens lemma and the cap-one operator modulus

Candidate full-solution packet for arXiv:1704.07095, Section 5

11 August 2026

Status: candidate full solution, likely valid, subject to human review. For every real Banach space X and $0 < \varepsilon < 1$,

$$\Phi^S(X, \mathbb{R}, \varepsilon) \leq \Phi(X, \mathbb{R}, \varepsilon) \leq \min\{\sqrt{2\varepsilon}, 1\}.$$

This answers the printed scalar problem affirmatively and strengthens it to the nonspherical modulus. The same cap-one estimate holds for every range space having property β with parameter zero.

Idea of the proof

The source's scalar problem [1] allows the approximating functional to attain either 1 or -1. Encode those two choices by the two boundary components of the lens $B_X \cap (x + B_X)$. A nearby functional supporting the lens splits into a nonnegative combination of the two ball normals. If the first coefficient dominates, its positive norming pair is close; if the second dominates, the translated-ball normal yields a close pair attaining -1. This gives the universal cap one. For a property-β range with parameter zero, a rank-one correction changes only one coordinate of the norming family and lifts the scalar construction to operators.

1 The geometric lemma

For a closed convex set $D \subset X$ and $u \in D$, write

$$N_D(u) = \{q \in X^* : q(v - u) \leq 0 \text{ for every } v \in D\}$$

for its normal cone. We use two standard convex-analytic facts. First, the Bishop–Phelps theorem says that the functionals attaining their supremum on a nonempty closed bounded convex set are dense in X^* [2]. Second, if closed convex sets A, D have intersecting interiors, then the interior-point normal-cone rule gives

$$N_{A\cap D}(u) = N_A(u) + N_D(u) \qquad (u \in A \cap D). \tag{1}$$

This is the normal-cone form of the Moreau–Rockafellar sum rule [3].

Lemma 1 (absolute-norming lens lemma)**.** *Let X be a real Banach space. For every $x \in B_X$ and $f \in B_{X^*}$,*

$$\inf_{\substack{y\in S_X,\ g\in S_{X^*} \\ |g(y)|=1}} \max\{\|x - y\|, \|f - g\|\} \leq 1.$$

Equivalently, for every $\delta > 1$ there are $y \in S_X$ and $g \in S_{X^*}$ with $|g(y)| = 1$ and both distances smaller than δ.

Proof. Put
$$C = B_X \cap (x + B_X).$$
The point $x/2$ is an interior point of both balls, so the qualification in (1) holds. Fix $\eta > 0$. By Bishop–Phelps choose a nonzero $p \in X^*$ with $\|p - f\| < \eta$ which attains its supremum on C, say at $u \in C$. When $f = 0$, nonzeroness can be ensured by first perturbing zero to a functional of norm smaller than $\eta/2$ and then applying the density theorem closely enough. Set

$$\varphi = \frac{p}{\max\{1, \|p\|\}}.$$

Then $\varphi \neq 0$, $\|\varphi\| \leq 1$, $\varphi \in N_C(u)$, and

$$\|\varphi - f\| < 2\eta. \tag{2}$$

Indeed, when $\|p\| > 1$ the extra normalization error is $\|p\| - 1 \leq \|p - f\|$.

The sum rule and the elementary normal cones of the two balls give

$$\varphi = \alpha g + \beta h, \tag{3}$$

where $\alpha, \beta \geq 0$; if $\alpha > 0$, then $u \in S_X$, $g \in S_{X^*}$, and $g(u) = 1$; if $\beta > 0$, then $u - x \in S_X$, $h \in S_{X^*}$, and $h(u - x) = 1$. At least one coefficient is positive.

If $\beta = 0$, then $\alpha = \|\varphi\| \leq 1$, and (u, g) is an absolute-norming pair satisfying

$$\|x - u\| \leq 1, \qquad \|\varphi - g\| = 1 - \alpha \leq 1.$$

If $\alpha = 0$, put $v = x - u$. Then $v \in S_X$, $h(v) = -1$, and

$$\|x - v\| = \|u\| \leq 1, \qquad \|\varphi - h\| = 1 - \beta \leq 1.$$

It remains to suppose that $\alpha, \beta > 0$. Put again $v = x - u$. Now $u, v \in S_X$, $x = u + v$, $g(u) = 1$, and $h(v) = -1$. Hence

$$g(v) = g(x) - 1 \leq \|x\| - 1 \leq 0, \qquad h(u) = h(x) + 1 \geq 1 - \|x\| \geq 0.$$

Evaluating (3) at u and v, using $\|\varphi\| \leq 1$, yields

$$\alpha + \beta h(u) = \varphi(u) \leq 1, \qquad \alpha g(v) - \beta = \varphi(v) \geq -1.$$

In particular, $\alpha \leq 1$ and $\beta \leq 1 + \alpha g(v) \leq 1$. If $\alpha \geq \beta$, select (u, g); then

$$\|x - u\| \leq 1, \qquad \|\varphi - g\| \leq (1 - \alpha) + \beta \leq 1.$$

If $\beta \geq \alpha$, select (v, h); then

$$\|x - v\| \leq 1, \qquad \|\varphi - h\| \leq \alpha + (1 - \beta) \leq 1.$$

Thus one of the two pairs is within one of (x, φ). By (2), it is within $1 + 2\eta$ of (x, f). Letting $\eta \downarrow 0$ proves the lemma. $\square$

2 The scalar problem is solved

We use the operator moduli exactly as in the source: the input pair is in $B_X \times B_{\mathcal{L}(X,Y)}$ for Φ, in $S_X \times S_{\mathcal{L}(X,Y)}$ for Φ^S, and the approximating operator has norm one and attains its norm at a unit vector.

Theorem 1. *For every real Banach space X and every $0 < \varepsilon < 1$,*

$$\Phi^S(X, \mathbb{R}, \varepsilon) \le \Phi(X, \mathbb{R}, \varepsilon) \le \min\{\sqrt{2\varepsilon}, 1\}.$$

Proof. Identify a scalar operator with $f \in B_{X^*}$. Lemma 1 says that every $(x, f) \in B_X \times B_{X^*}$ can be approximated, to every distance strictly larger than one, by $y \in S_X$ and $g \in S_{X^*}$ with $|g(y)| = 1$. Thus $\Phi(X, \mathbb{R}, \varepsilon) \le 1$; the near-attainment hypothesis is not even needed for this half.

The classical Bishop–Phelps–Bollobás estimate, or equivalently the source's property-β theorem at parameter zero, gives $\Phi(X, \mathbb{R}, \varepsilon) \le \sqrt{2\varepsilon}$. Taking the smaller bound and using $\Phi^S \le \Phi$ proves the claim. □

This is stronger than the printed question in two ways: it bounds the nonspherical modulus, and the cap-one part holds for every input pair rather than only for pairs nearly attaining a norm. Together with the source's sharp lower bound, it also recovers

$$\Phi^S(\ell_1^{(2)}, \mathbb{R}, \varepsilon) = \Phi(\ell_1^{(2)}, \mathbb{R}, \varepsilon) = \min\{\sqrt{2\varepsilon}, 1\}.$$

3 Upgrade: all property-β ranges with parameter zero

Recall that $\beta(Y) = 0$ means that there are $(y_\gamma, y_\gamma^*)_{\gamma\in\Gamma} \subset S_Y \times S_{Y^*}$ such that

$$y_\gamma^*(y_\gamma) = 1, \qquad y_\gamma^*(y_\nu) = 0 \ (\gamma \ne \nu), \qquad \|w\| = \sup_\gamma |y_\gamma^*(w)| \ (w \in Y).$$

Theorem 2. *If X, Y are real Banach spaces and $\beta(Y) = 0$, then for every $0 < \varepsilon < 1$,*

$$\Phi^S(X, Y, \varepsilon) \le \Phi(X, Y, \varepsilon) \le \min\{\sqrt{2\varepsilon}, 1\}.$$

Proof. Only the cap one is new. Let $x \in B_X$, $T \in B_{\mathcal{L}(X,Y)}$, and $\|Tx\| > 1 - \varepsilon$. Choose γ_0 with $|y_{\gamma_0}^*(Tx)| > 1 - \varepsilon$ and put $f = T^* y_{\gamma_0}^* \in B_{X^*}$. For any $\delta > 1$, the lens lemma gives $z \in S_X$ and $g \in S_{X^*}$ such that

$$|g(z)| = 1, \qquad \|x - z\| < \delta, \qquad \|f - g\| < \delta.$$

Define

$$F(t) = T(t) + (g(t) - f(t))y_{\gamma_0} \qquad (t \in X).$$

Orthogonality of the property-β family gives

$$F^* y_{\gamma_0}^* = g, \qquad F^* y_\gamma^* = T^* y_\gamma^* \quad (\gamma \ne \gamma_0).$$

Since the family (y_γ^*) is norming,

$$\|F\| = \sup_\gamma \|F^* y_\gamma^*\| = 1.$$

Moreover $\|Fz\| \ge |y_{\gamma_0}^*(Fz)| = |g(z)| = 1$, so F attains its norm at z, and

$$\|F - T\| = \|g - f\| < \delta.$$

Thus $\Phi(X, Y, \varepsilon) \le 1$. The $\sqrt{2\varepsilon}$ bound is Theorem 2.1 of the source with $\rho = 0$. □

This settles the source's second, more general cap-one question only in the endpoint case $\rho = 0$. The argument uses the exact vanishing of all off-diagonal coefficients and does not prove the cap for $0 < \rho < 1$.

4 Novelty and verification boundary

On 11 August 2026, exact searches of the run registry and all cheap indexes, the locally parsed arXiv corpus, and bounded web searches found the source but no later answer to its scalar question. A later-title and citation scan also found no theorem matching the cap-one claim. This is a bounded novelty check, not an exhaustive priority claim.

The substantive external inputs are the Bishop–Phelps density theorem, the interior-point normal-cone sum rule, and the source's already-proved $\sqrt{2\varepsilon}$ estimate. The cap-one argument itself is independent of the near-attainment condition and is given in full above.

A Source-page evidence

Figure 1 reproduces the source page containing both open questions.

References

[1] V. Kadets and M. Soloviova, *Quantitative version of the Bishop–Phelps–Bollobás theorem for operators with values in a space with the property β*, arXiv:1704.07095 [math.FA], 2017.

[2] E. Bishop and R. R. Phelps, *A proof that every Banach space is subreflexive*, Bull. Amer. Math. Soc. 67 (1961), 97–98.

[3] R. R. Phelps, *Convex Functions, Monotone Operators and Differentiability*, Lecture Notes in Mathematics 1364, 2nd ed., Springer, 1993.

5. AN OPEN PROBLEM

Problem 5.1. *Is it true, that* $\Phi^S(X, \mathbb{R}, \varepsilon) \leqslant \min\{\sqrt{2\varepsilon}, 1\}$ *for all real Banach spaces* X*?*

In order to explain what do we mean, recall that for the original Bishop-Phelps-Bollobás modulus the estimation

$$\Phi^S_X(\varepsilon) \leqslant \sqrt{2\varepsilon} \tag{5.1}$$

holds true for all X. In other words, for every $(x, x^*) \in S_X \times S_{X^*}$ with $x^*(x) > 1 - \varepsilon$, there is $(y, y^*) \in S_X \times S_{X^*}$ with $y^*(y) = 1$ such that $\max\{\|x - y\| < \sqrt{2\varepsilon}, \|x^* - y^*\|\} < \sqrt{2\varepsilon}$.

When we take $Y = \mathbb{R}$ in the definition of $\Phi^S(X, Y, \varepsilon)$, the only difference with $\Phi^S_X(\varepsilon)$ is that by attaining norm we mean $|y^*(y)| = 1$, instead of $y^*(y) = 1$. So, in the case of $\Phi^S(X, \mathbb{R}, \varepsilon)$ we have more possibilities to approximate $(x, x^*) \in S_X \times S_{X^*}$ with $x^*(x) > 1 - \varepsilon$:

$$(y, y^*) \in S_X \times S_{X^*} \text{ with } y^*(y) = 1 \text{ or } y^*(y) = -1.$$

Estimation (5.1) is sharp for the two-dimensional real ℓ_1 space:

$$\Phi^S_{\ell_1^{(2)}}(\varepsilon) = \sqrt{2\varepsilon}, \tag{5.2}$$

but, as we have shown in Theorem 3.4

$$\Phi^S(\ell_1^{(2)}, \mathbb{R}, \varepsilon) = \min\{\sqrt{2\varepsilon}, 1\}. \tag{5.3}$$

Estimations (5.2) and (5.3) coincide for $\varepsilon \in (0, 1/2)$, but for bigger values of ε there is a significant difference. We do not know whether the inequality $\Phi^S(X, \mathbb{R}, \varepsilon) \leqslant \min\{\sqrt{2\varepsilon}, 1\}$ holds true for all X.

Moreover, in all examples that we considered we always were able to estimate $\Phi^S(X, Y, \varepsilon)$ from above by 1. So, we don't know whether the statement of Theorem 2.1 can be improved to

$$\Phi^S(X, Y, \varepsilon) \leqslant \min\left\{\sqrt{2\varepsilon}\sqrt{\frac{1+\rho}{1-\rho}},\ 1\right\}.$$

REFERENCES

[1] M. D. ACOSTA, R. M. ARON, D. GARCÍA AND M. MAESTRE, The Bishop-Phelps-Bollobás Theorem for operators, *J. Funct. Anal.* **254** (2008), 2780–2799.

[2] M. D. ACOSTA, J. BECERRA GUERRERO, D. GARCÍA, S. K. KIM, AND M. MAESTRE, Bishop-Phelps-Bollobás property for certain spaces of operators, J. Math. Anal. Appl. **414** (2014), No. 2, 532–545.

[3] BISHOP, E., PHELPS, R.R.: A proof that every Banach space is subreflexive, *Bull. Amer. Math. Soc* **67** (1961), 97–98.

[4] BOLLOBÁS, B.: An extension to the theorem of Bishop and Phelps, *Bull. London Math. Soc.* **2** (1970), 181–182.

[5] CHICA, M., KADETS, V., MARTÍN, M., MORENO-PULIDO, S., RAMBLA-BARRENO, F.: *Bishop-Phelps-Bollobás moduli of a Banach space*, *J. Math. Anal. Appl.* **412** (2014), no. 2, 697–719.

[6] CHICA, M., KADETS, V., MARTÍN, M., MERÍ, J., SOLOVIOVA, M.: Two refinements of the Bishop-Phelps-Bollobás modulus, *Banach J. Math. Anal.* **9** (2015), no. 4, 296–315.

[7] CHICA, M., KADETS, V., MARTÍN, M., MERÍ, J.: *Further Properties of the Bishop-Phelps-Bollobás Moduli*, Mediterranean Journal of Mathematics 13(5):3173-3183 (2016)

[8] DIESTEL, J.: *Geometry of Banach spaces*, Lecture notes in Math. **485**, Springer-Verlag, Berlin, 1975.

[9] R. C. JAMES Uniformly non-square Banach spaces, Ann. Math. (2) **80** (1964), 542–550.

[10] PARTINGTON J.: *Norm attaining operators*, *Israel J. Math.* 43 (1982), 273-276.

[11] V. KADETS, M. SOLOVIOVA: *A modified Bishop-Phelps-Bollobás theorem and its sharpness*, Mat. Stud. **44** (2015), 84–88.

[12] J. LINDENSTRAUSS: *On operators which attain their norm*, Israel J. Math. **1** (1963), 139–148.

[13] PHELPS, R. R.: Support Cones in Banach Spaces and Their Applications, *Adv. Math.* **13** (1974), 1–19.

Figure 1: The source's final printed page, including the scalar question and the property-β cap-one question.